

MAM Solution Approaches · Independent Review Record

Review method: Each set of notes is reviewed by an independent subagent. The reviewer did not take part in writing, works with fresh context, and only reads, does not modify. Evidence checked includes: the original "Specific behaviours" text in each question's marking key in *evidence/*, the full text of the 2016–2025 exam reports, the PDF pages of the original papers and keys, with independent verification using sympy/scipy.

Handling principles: The author reviews the review comments one by one. All high-severity comments were only changed after going back to the original paper wording and verifying; everything that could be verified by calculation was checked. Items marked "doubtful" by the reviewer were changed only when supporting evidence could be found.

Outcome categories: Accepted = changed in line with the comment; Partially accepted = wording changed but the suggestion was not followed completely; Rejected = not changed, with reasons given.

01 Differentiation and its applications (15 items: accepted 15)

#	Severity	Issue	Action
1	High	"If you do not verify the extremum you get at most half marks" is an invention by the author; the 2024 report does not say this; in 2024A-Q12(c), verification accounts for only 1 of the 4 marks	Accepted: changed to "verification is usually worth 1 mark on its own"
2	Medium	"2019 report: one of the worst on the paper, typical error $h'(1)$ " does not match the original report	Accepted: changed to the report's exact words, with $h'(1)$ noted as an author reminder, and added 2020 report Q5(c)
3	Medium	"Verification is guaranteed marks and must not be omitted" overstates it: the wording of 2025A-Q12(b) and 2025A-Q14(c) states that verification is not required	Accepted
4	Low	"Missing one type of label loses one mark" is not the report's exact wording; the question number format was wrong	Accepted: changed to the actual 4 marking points for 2020F-Q7(d)
5	Medium	"Horizontal point of inflection ($f'=f''=0$)" may be mistaken for a test condition; counterexample x^4	Accepted
6	Low	"Writing the two-term structure earns marks" is inaccurate; the key requires at least one term to be correct	Accepted
7	Low	"2023 report: the conclusion was judged invalid" is not the report's exact wording	Accepted
8	Low	"The report warns not to write it" cannot be found in the MAM reports	Accepted: changed to the actual statements in the 2018, 2019 and 2025 reports
9	Low	2025F-Q3 is not related rates	Accepted: moved out of Pattern 8
10	Low	Pattern 3 incorrectly cites 2017F-Q6(b)(ii); omitted 2018F-Q3(b) and 2019F-Q7(b)	Accepted
11	Low	The reason for deducing $P''(6)=0$ in 2018A-Q15 was not clearly written	Accepted: added "continues to increase afterwards $\Rightarrow$ horizontal point of inflection", and checked $a=-18$, $b=108$
12	Low	"Each condition is worth 1 mark" is an overgeneralisation	Accepted
13	Low	"The derivative does not exist at a turning point" is ambiguous wording	Accepted: changed to "corner point (sharp corner)"
14	Low	The numbering explanation did not explain 2016S (sample paper)	Accepted
15	Low	The expressions " $x^2e^x \rightarrow xe^x(x+2)$ " and " $A''(x)<0$ " were unclear	Accepted

03 Rectilinear motion (11 items: accepted 11)

#	Severity	Issue	Action
1	High	"When $a = 0$, velocity is at an extremum" is an overgeneralisation: $v = t^3$ is a counterexample; in the constant-velocity section of 2019A-Q12(c), a is always 0; endpoints can also give an extremum	Accepted
2	Medium	In "hits 0 and turns back", the meaning of "turns back" is exactly the opposite	Accepted
3	Medium	The piecewise integral was written as "+ −", opposite to the cited 2023A-Q8(c) (that question is negative first, then positive)	Accepted: after the author verified $v = -2t \sin t$ by calculation and confirmed, changed to "add a negative sign for the $v < 0$ section" and gave the working for that question
4	Medium	Report years were wrong: 2021 was not listed as a major weakness; 2018 and 2023 were omitted	Accepted
5	Medium	The key for 2018A-Q11(c) treats "decelerating 0.5" as $a = -0.5$, which conflicts with the same-sign/opposite-sign rule	Accepted: the author checked the original question wording and added a note as a reminder
6	Medium	"Returns to the origin" and "returns to the starting point" were used interchangeably	Accepted
7	Low-Medium	" $v < 0$ or compare endpoints": questions using calculus must use v or a sign test	Accepted
8	Low	"Leaves the left side of the screen" is ambiguous	Accepted: changed to "to the right"
9	Low	The graphing requirements for 2025F-Q6(d) were written imprecisely	Accepted
10	Low	"Divide by the inner coefficient" is easy to miscalculate	Accepted: added "i.e. multiply by 3"
11	Low	Omitted citations of 2017A-Q20(c)(d) and 2019A-Q12	Accepted

05 Discrete random variables (13 items: accepted 13)

#	Severity	Issue	Action
1	High	2018A-Q12(f) omitted the "additional \$1 charge"; the new mean should be $1.2\mu + 1$	Accepted: the author confirmed after checking the original question
2	Medium	"4 decimal places, otherwise marks are deducted": the report does not say marks are deducted; the key for 2019A-Q18(c) requires at least 5 places	Accepted: the author verified $P \approx 3.0 \times 10^{-5}$ by calculation
3	Medium	$Y = 3 - X$ is negative when $X = 4$	Accepted: changed to $\max(0, 3 - X)$
4	Medium	"Every year" is untrue; in fact it is 6 out of 10 years	Accepted
5	Medium (doubtful)	Assumptions question: the key only gives "independent" and "same p "; "two outcomes" and " n fixed" are given in the question	Accepted: supported by the original key
6	Medium	"binomial and parameters worth 1 mark separately" is ambiguous	Accepted
7	Medium	Pattern 5 incorrectly cites 2022A-Q12 (there is no binomial sub-question); the single-item probability in 2023A-Q13(b) comes from a uniform distribution	Accepted: deleted the former, replaced it with 2021A-Q8(d), and added a note to the latter
8	Low	$a = \pm 15/22$, with no explanation of why the positive value is taken	Accepted
9	Low	"2020 report Q1" is not labelled F	Accepted
10	Low	2016S-F4 is cited, but it is not stated that it is not among the 28 questions	Accepted
11	Low	The marking points for 2017A-Q13 omitted the 2 marks for (d)	Accepted

12	Low	Pattern 6 incorrectly classified 2021A-Q10(b) and 2023A-Q13(a) as "use binomial"	Accepted
13	Low	"Probability expression worth 1 mark separately" is an overgeneralisation	Accepted: changed to "most questions"

06 Continuous random variables and the normal distribution (13 items + 1 doubtful: accepted 13, partially accepted 1)

#	Severity	Issue	Action
1	High	Applying $(b-a)^2/12$ directly for the variance of a uniform distribution loses marks: the key awards marks for "pdf and domain" and "integral expression", and this formula is not on the formula sheet either	Accepted: changed to writing an integral for the answer, with the formula used only for checking
2	Medium	"Keep 4 decimal places" is an overgeneralisation; some questions specify 3 places or exact	Accepted
3	Medium	It reminds students to distinguish cdf, but does not teach the cdf question type (2023A-Q11(d)(e))	Accepted: added to Pattern 2
4	Medium	In 2021F-Q6(b)(ii), comparing using σ does not lead to the conclusion; the key uses area	Accepted: changed to the key's reasoning, and added a strict bound (author verified 0.383)
5	Low	"Designed according to the 68-95-99.7 rule" is inaccurate	Accepted
6	Low	"2.9 h $\rightarrow$ 2:54 pm" is not data from the actual exam question	Accepted: changed to 3.8 h $\rightarrow$ 3:48 pm from 2023A-Q11(c)
7	Low	2025F-Q4 is "Show that k", not "Determine"	Accepted: both questions cited
8	Low	"Probability expression worth 1 mark separately" is an overgeneralisation	Accepted
9	Low	Unit conversion omitted the variance, and also reverse-finding a and b	Accepted
10	Low	The unit of variance was omitted	Accepted
11	Low	The histogram question type omitted frequency-to-probability conversion, cumulative tables, and estimating the expected value from class midpoints	Accepted
12	Low	The notation for 2016S-A12 is inconsistent	Accepted
13	Low	Quotation marks were added around an identifying sentence, making it look like the original wording when it was actually a paraphrase	Accepted
Doubtful	—	"2021 report: many people did not write uniform" goes beyond the original report wording	Partially accepted: changed to the report's actual statement plus the key answer

02 Integration and its applications (16 items: accepted 14, partially accepted 2)

#	Severity	Issue	Action
1	Medium	"FTC is named and examined every year" is untrue (it was not examined in 2019, 2020 or 2024, and many years' reports do not name it)	Accepted: changed to "high-frequency + named many times" and listed the years
2	Medium	2021A-Q9 does not involve integration	Accepted: moved out of Pattern 8 and a note added
3	Medium	"The 3 marking points in the key are exactly these three steps" is an overgeneralisation	Accepted
4	Medium	The rectangle estimate only covered increasing functions; 2021F-Q7 is decreasing ($1/x$)	Accepted

5	Medium	"Set the cross-sectional area equal to the volume" is dimensionally incorrect	Accepted: changed to "cross-sectional area $\times$ length = volume"
6	Low	2023F-Q5(b) is not of the form $\int f'$	Accepted: replaced with 2023F-Q2(a)(iii), with Q5(b) listed separately
7	Low	The A(x) rule omitted "f changes from negative to positive $\rightarrow$ local minimum"	Accepted
8	Low-Medium	The question type involving direct evaluation of a definite integral was omitted, and 2020F-Q3 was not classified	Accepted: added to Pattern 1 (2020F-Q3 was already cited in Pattern 5)
9	Low (doubtful)	"The report says many people lost marks here" is unsupported	Accepted: changed to the actual statement in the 2017 report
10	Low	"Use CAS to calculate the exact value" does not match the CF paper	Partially accepted: changed to "find the function then integrate", with the trapezium retained
11	Low	The constant c appears out of nowhere	Accepted
12	Low	$d/dx(x \sin 3x)$ should be $2x \sin 3x$	Accepted
13	Low	2023F-Q2(c)(iii) is not an area question	Accepted
14	Low	"Crossing the x-axis requires splitting into intervals" is an overgeneralisation	Accepted
15	Low	Missing $n \neq -1$; WACE convention is to write $\ln f(x)$	Accepted
16	Low	The explanation and question numbering format for the 2016 sample paper	Partially accepted: an explanation was added at the beginning, but the question numbering format was kept as 2016S-F1 and not changed

04 Exponentials and Logarithms (12 items: accepted 11, partially accepted 1)

#	Severity	Issue	Resolution
1	Medium	Summary of correspondence between report years and weak areas is inaccurate	Accepted: list item by item by year
2	Medium	The 2023 report refers to readings (2023F-Q4(c)), not ratios; "cannot be subtracted" conflicts with the key	Accepted
3	Medium	The value in "9 am + 4.7 h $\Rightarrow$ 1:42 pm" is wrong; 2018A-Q9(d) is actually 9.40 h $\Rightarrow$ 6:24 pm	Accepted: after the editor checked $\ln 1.6/0.05 = 9.40$, corrected it and added rounding down for latest
4	Medium	The ratio formula only had /10; 2016S-A13 ($k=20$), 2016A-Q12 ($k=1$), and 2023F-Q4 ($k=2$) differ	Accepted: changed to the general formula $L = k \cdot \log$
5	Medium	"Write the rate of decrease as a positive number" is overgeneralised; the key for 2018A-Q9(c) gives a negative number	Accepted
6	Medium-high	Missing the "calculus of $e^x / \ln x$ " pattern; 13 questions are unassigned	Partially accepted: added cross-reference Pattern 7; methods are in the differentiation and integration notes to avoid duplication
7	Low	2016 sample paper not specified	Accepted
8	Low	"Almost every question's first step is converting between forms" is overgeneralised	Accepted
9	Low	Pattern 1 incorrectly cites past exam questions 2016F-Q1 and 2025F-Q7(a)	Accepted
10	Low	Missing context-based rounding rules	Accepted
11	Low	WACE key often writes $\ln f(x)$	Accepted
12	Low	"Dashed lines" are not a key requirement	Accepted

07 Sample Proportions and Confidence Intervals (11 items: accepted 11)

#	Severity	Issue	Resolution
1	High	The $\hat{p}$ rule for minimum sample size is reversed: the key still uses 0.5 when $\hat{p}$ is known (2023A-Q7(c), 2024A-Q10(f), 2025A-Q16(e); the 2023 report specifically warns about this)	Accepted: the editor checked the question stems for all three questions and corrected it; under the old rule, 2023A-Q7(c) would give 2263 instead of 2401. The question-writing skill was also corrected
2	Medium	The 2023 report's warning was paired with a 2024 example	Accepted
3	Medium	2025A-Q15(f) is a "prove" question, the answer is No, and it was categorised incorrectly	Accepted: listed separately as "distinguishing the two question types"
4	Medium	The example combined time and location into one bias, violating the rule on the same page	Accepted: split into two
5	Medium	Missing binomial distribution assumption questions (2020A-Q12(h), 2021A-Q11(g))	Accepted
6	Low	"2–4 marks per year / 2 marks per source" is overgeneralised	Accepted
7	Low	2024A-Q10(e) belongs to Pattern 5; Pattern 5 is missing a list of past exam questions	Accepted
8	Low	The calculation with 9603.6 and 1.96 is inconsistent	Accepted
9	Low	"Switch to the binomial distribution" was mistakenly written as a remedy for (a)	Accepted
10	Low	"Random" is not a key requirement for 2016A-Q10(a)	Accepted
11	Low (doubtful)	Reverse-solving for the confidence level omitted 2023A-Q12(c)(d)	Accepted

Summary (7 MAM papers): 92 comments in total (including 1 "doubtful"), 88 accepted, 4 partially accepted, 0 rejected. There are 5 high-severity items; all were revised after checking against the original exam paper wording.

Category Corrections After User Feedback (2026-10-09)

After a user pointed out that the vector question bank contained unrelated questions, all tags were keyword-reviewed, and the key was checked item by item:

- MAM 2021A-Q9 removed the "Integration" tag (the whole question only involves solving equations, differentiation, and finding extrema; there is no integration; the integration notes reviewer also pointed this out).
- MAM 2024A-Q10 removed the "Discrete Random Variable" tag (the whole question has no binomial distribution).